TASK 03

POLYNOMIAL REGRESSION MODEL

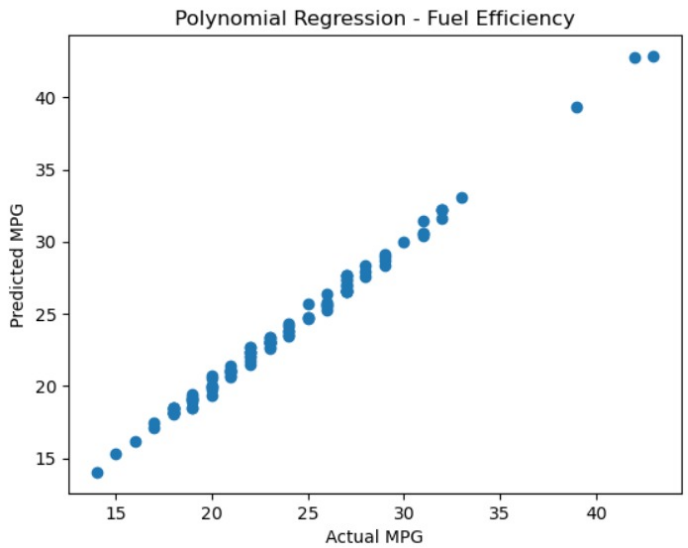
A car manufacturer wants to predict the fuel efficiency (miles per gallon, MPG) of its vehicles based on engine size, weight, and horsepower. You are tasked with building a Polynomial Regression model to capture non-linear relationships between features and MPG.

CODE AND OUTPUT :

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

df = pd.read_csv("car_data.csv")

df = df.dropna()

X = df[['city_mpg', 'highway_mpg', 'cylinders', 'displacement', 'year']]
y = df['combination_mpg']

X = X.values
y = y.values.reshape(-1,1)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

poly_features = PolynomialFeatures(degree=2)
X_poly_train = poly_features.fit_transform(X_train)
X_poly_test = poly_features.transform(X_test)

poly_model = LinearRegression()
poly_model.fit(X_poly_train, y_train)

y_poly_pred = poly_model.predict(X_poly_test)
```

```python
plt.scatter(y_test, y_poly_pred)
plt.xlabel("Actual MPG")
plt.ylabel("Predicted MPG")
plt.title("Polynomial Regression - Fuel Efficiency")
plt.show()

print(f"Coefficients: {poly_model.coef_}")
print(f"Intercept: {poly_model.intercept_}")

from sklearn.metrics import mean_squared_error, r2_score

mse = mean_squared_error(y_test, y_poly_pred)
r2 = r2_score(y_test, y_poly_pred)

print(f"Mean Squared Error: {mse}")
print(f"R^2 Score: {r2}")
```

OUTPUT:




```
Actual MPG
Coefficients: [[ 0.00000000e+00  3.86714032e-01  7.32444829e+00  1.63612101e+01
   4.28950496e+01  1.58370274e+01 -1.03052760e-02  1.96629741e-02
   4.14898661e-02 -4.66958461e-02  4.59282175e-05 -9.82420190e-03
  -1.19886581e-02 -1.23369347e-03 -3.36846066e-03  9.00358508e-02
  -2.14160626e-01 -8.53651727e-03  3.54501255e-02 -2.03097890e-02
  -3.87713596e-03]]
Intercept: [-16170.93368622]
Mean Squared Error: 0.12816636865939365
R^2 Score: 0.9952220340974728
```